

**RAMCO INSTITUTE OF TECHNOLOGY**  
**DERPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING**  
**Academic Year: 2018 - 2019 (Even Semester)**  
**Innovative Practices Description**

|                                        |                                                                                     |
|----------------------------------------|-------------------------------------------------------------------------------------|
| <b>Degree, Semester &amp; Branch</b>   | : VIII Semester B.E. EEE                                                            |
| <b>Course Code &amp; Title</b>         | : EE6009 Power Electronics for Renewable energy Systems                             |
| <b>Name of the Faculty member</b>      | : Mrs. S. Jeyanthi                                                                  |
| <b>Name of the Topic</b>               | : Fundamentals of solar cell, Connections of solar panels, importance of inverters. |
| <b>Name of the Innovative Practice</b> | : Field Visit                                                                       |
| <b>Date &amp; Duration</b>             | : 11.03.2019 & 50 minutes                                                           |
| <b>ICT Tool Used</b>                   | : -                                                                                 |

**Description:**

A field visit was arranged for final year EEE students on 11.03.2019 at **RIT – Solar Plant** from 03.20 PM to 04.10 PM. The students were accompanied by two faculty members, Mrs. S. Jeyanthi AP/EEE and Mr.D.Karthik Prabhu, AP(SG)/EEE. The students were separated into 2 batches. The following points were explained to the students:

- Layout and installation of solar plant.
- Solar plant installed in college premises consists of totally 42 panels. Out of which 14 panels are connected in series.
- Each panel produces 30 V DC, 250 W.
- It is a grid connected solar power plant and it has the capability to generate a maximum of 40 units per day.
- The material used for making the solar is polycrystalline material.
- The inverters connection to the solar plant.
- Standalone operation and Grid integration of solar system.

The visit came to an end at 04.10 PM. Students were also made to clarify their doubts regarding solar power plan.

**Goals (Learning Outcomes):**

- Students will be able to explain about the types of connection of solar pannel .
- Students will be able to expose to the material used for solar cell.

**Use of appropriate methods:**

**Field visit** was chosen to explore the students about the material used in solar cell, inverters, types of connection of solar panel, standalone operation of solar system and grid integrated solar system.

**Effective presentation**

**Details of the Implementation:**

- After teaching the concept of standalone operation of solar system and grid integrated solar system, students were took to the RIT-Solar plant.
- At the end of the visit, students were able to understand the different operation of solar system and connection of solar plant.



### **A view of the implementation of activity:**

#### **Significance of Results:**

#### **Assessment of Effectiveness/Success of the Activity:**

- The assessment of effectiveness of the activity was felt when told most of the points.
- During the visit, I understood that there should follow the sequence while mention the components in a substation and finally I insisted the components sequentially.

#### **Reflective Critique:**

- **Benefits:**
  - Students were understood clearly by visualizing the connection of solar plant through the visit.
- **Challenges:**
  - Time utilization for conducting activity.

#### **Changes required for future (if required):**

- In future, increase the time duration for the field visit.

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 | PSO4 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|------|
| CO1 | 3   | 2   | 2   | 1   | 1   | -   | -   | -   | -   | -    | -    | 1    | 2    | 1    | 2    | -    |
| CO3 | 3   | 2   | 1   | 1   | 1   | -   | -   | -   | -   | -    | -    | 1    | 2    | 1    | 2    | -    |
| CO4 | 3   | 2   | 2   | 1   | 1   | -   | -   | -   | -   | -    | -    | 1    | 2    | 2    | 2    | -    |
| CO5 | 3   | 2   | 2   | 1   | 1   | -   | -   | -   | -   | -    | -    | 1    | 2    | 2    | 2    | -    |

- CO1 Explore the environmental aspects of electrical energy generation and utilization.
- CO3 Explain the different power converters namely DC to AC, AC to DC, DC to DC and AC to AC converters for renewable energy sources.
- CO4 Compare and analyze the various operating modes of wind electrical generators and solar energy systems.
- CO5 Analyze the different type of Hybrid renewable energy system.

**References:**

1. S. N. Bhadra, D.Kastha, S.Banerjee, "Wind Electrical Systems", Oxford University Press, 2005.
2. B.H.Khan, Non-conventional Energy sources Tata McGraw-hill Publishing Company, New Delhi, 2009.